A Controlled Study of Mixing Supervision and Latent Arithmetic in Audio Autoencoders

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***Abstract*—Explicit mixing supervision aims to improve audio composition through latent arithmetic, but common evaluations measure different properties of an autoencoder. We investigate these effects through same-start fine-tuning comparisons in GAN audio autoencoders. Decode-mixing supervision improves ground-truth convex mixing with little change in reconstruction SI-SDR. An independent held-out corpus confirms a $1.73\,\mathbf{dB}$ mixing improvement at $7.66\times$ compression and $1.33\,\mathbf{dB}$ for a combined recipe at $15.3\times$. Discriminator supervision, however, increases decoder consistency while reducing ground-truth mixing fidelity. For source subtraction, correcting the latent origin removes 87–95% of the raw between-model advantage on 240 held-out recordings. These results distinguish waveform mixing improvements from decoder agreement and from origin-dependent subtraction effects. Findings are conditional on single training runs; held-out evaluation uses frozen checkpoints that were selected on the original evaluation data.**

***Index Terms*—audio autoencoders, latent arithmetic, evaluation methodology, representation learning, controlled ablation**

I. Introduction

Audio mixing is linear on waveforms, $\mathrm{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$, which makes latent arithmetic an attractive interface to a learned representation. For an encoder f and decoder g, the property usually asked for is

$$g(\alpha f(x_1) + (1-\alpha) f(x_2)) \approx \alpha x_1 + (1-\alpha)x_2, \qquad (1)$$

and a common companion probe is stem removal by subtraction, $g(f(x_{\mathrm{mix}})-f(x_{\mathrm{stem}})) \approx x_{\mathrm{res}}$. Where these hold, editing and composition happen in the latent domain without a dedicated separation or synthesis model, a prerequisite for latent-diffusion systems [1], [2] that compose independently modeled sources. Standard autoencoders do not satisfy Eq. 1: their objectives reward reconstruction fidelity, not algebraic structure, and in quantized codecs (DAC [3], EnCodec [4]) an interpolated point is not a codebook entry.

The supervision studied here is not ours. Torres et al. [5] induce linearity in a consistency autoencoder by decoding the average of separately encoded latents against the true mix, a principle related to mixup [6]. We write that same supervision as an explicit loss on a waveform GAN autoencoder. The implementation differs from theirs in loss family (multi-resolution STFT, mel and a waveform ℓ_1 term rather than a consistency loss), in mixing coefficient ($\alpha \sim U[0,1]$ rather than 0.5), and in decoder class, so this is neither a new method nor an exact reproduction. What we contribute is the controlled comparison and the analysis of what the standard measurements actually report:

1) **Supervision comparison.** Same-start fine-tuning shows how decode-mixing supervision, an encoder-side mixing loss and decoder-only adaptation affect ground-truth mixing and reconstruction (Section IV-A, Section IV-B).
2) **Evaluation comparison.** Adding discriminator supervision on the mixed path improves agreement between the two decoded outputs while reducing ground-truth mixing fidelity, so the two measurements favour different models and are not interchangeable evidence (Section IV-C).
3) **Origin analysis.** A short argument shows that recentering the latent leaves reconstruction and every unit-sum combination pointwise unchanged while altering subtraction, and empirically the correction substantially reduces between-model subtraction differences, including on an independent corpus (Section IV-D).

II. Related Work

Neural audio codecs. DAC [3], EnCodec [4] and SoundStream [7] pair convolutional encoders with HiFi-GAN decoders [8] and multi-scale discriminators, optimizing reconstruction without latent structure; under their residual vector quantization an interpolated point is generally not a codebook entry. RAVE [9] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [10] encourage generic smoothness rather than a specific algebraic identity. Mixing-equivariance follows the symmetry-based view of representation learning [11], though convex mixing is an affine combination, not a group action. **Linear audio representations.** Music2Latent [12] is a consistency autoencoder with partial linearity as a byproduct, and Torres et al. [5] make it approximately linear by the supervision described above. Their additivity and separation scores compare two decodes, a protocol Section III-C shows to be sensitive to decoder sampling.

III. Method

A. Mixing supervision

Given a batch $\{x_i\}$ we form pairs (x_i, x_j) with a fixed-point-free permutation and draw $\alpha \sim U[0,1]$. With $\bar{x} = \alpha x_i + (1-\alpha)x_j$ and $\bar{z} = \alpha f(x_i) + (1-\alpha) f(x_j)$, the decode-mixing loss requires the decoded interpolation to match the true mix:

$$\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} = \mathcal{L}_{\mathrm{recon}}(g(\bar{z}), \bar{x}) + \mathcal{L}_{\text{wave-}\ell_1}(g(\bar{z}), \bar{x})\,, \qquad (2)$$

where $\mathcal{L}_{\mathrm{recon}}$ is the weighted multi-resolution STFT and mel loss also used for reconstruction and $\mathcal{L}_{\text{wave-}\ell_1}$ is the unweighted mean absolute error over all samples and batch elements. The reconstruction path carries no waveform term, so the two paths differ in objective. Gradients flow through g and into f via $\bar{z}$, so one term constrains both networks. We also study an encoder-only variant, $\mathcal{L}^{\mathrm{enc}}_{\mathrm{mix}} = \|f(\bar{x}) - \bar{z}\|^2$, and a variant that additionally applies the discriminator to the mixed path ("+disc"). The total objective is $\mathcal{L} = \mathcal{L}_{\mathrm{recon}} + \mathcal{L}_{\mathrm{GAN}} + \lambda\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}} + \lambda_{\mathrm{enc}}\mathcal{L}^{\mathrm{enc}}_{\mathrm{mix}} + \lambda_{\ell_2}\|z\|^2$ with λ=0.5, λ_{ℓ_2}=10^{-3}, equal multi-resolution STFT and mel weights, adversarial weight 0.5 and feature-matching weight 2; λ_{enc} is 0 except in the encoder-only runs, which sweep 5, 10 and 20. Training uses Adam at 10^{-4} (5×10^{-5} for the discriminator), β=$(0.8, 0.99)$, weight decay 10^{-3}, gradient clipping at 1.0, batch 32 with two accumulation steps, and 500 warmup steps.

B. The latent origin is a free coordinate

Proposition. For arbitrary f, g and $c = f(0)$, define $f_c(x) = f(x) - c$ and $g_c(z) = g(z + c)$. Then $g_c(f_c(x)) = g(f(x))$ for every x; for any coefficients with $\sum_i a_i = 1$, $g_c(\sum_i a_i f_c(x_i)) = g(\sum_i a_i f(x_i))$; and

$$g_c(f_c(x) - f_c(y)) = g(f(x) - f(y) + c)\,. \tag{3}$$

Proof. $f_c(x) + c = f(x)$ gives the first. For the second, $\sum_i a_i f_c(x_i) = \sum_i a_i f(x_i) - c$ because the coefficients sum to one, and g_c adds c back. For the third the coefficients sum to zero, so c cancels in the difference and g_c adds it inside g. $\square$

Reconstruction and every unit-sum combination are therefore pointwise unchanged by this relabelling, for arbitrary nonlinear networks, while subtraction is not. Any difference a subtraction score shows under it is a property of the coordinate system rather than of the autoencoder as a function. When f is affine, $f(x) - f(y) + c = f(x-y)$, so Eq. 3 decodes $g(f(x_{\mathrm{res}}))$, the model's own reconstruction of the residual, and we report the shortfall from that reference as the *linearity tax*. The proposition is general, but it concerns re-expressing a *frozen* model: it does not make training origin-invariant, and the latent ℓ_2 penalty in particular depends on the origin, so the objective can itself move $f(0)$. The magnitude of the effect in Section IV-D is an empirical finding about the evaluated checkpoints and corpora, and neither it nor improved subtraction quality follows from the algebra for arbitrary models.

C. Metrics

We report reconstruction SI-SDR$_{\mathrm{rec}}$ = $\mathrm{SISDR}(g(f(x)), x)$ and, at α=0.5, two views of Eq. 1: SI-SDR$_{\mathrm{lin}}$ = $\mathrm{SISDR}(g(\bar{z}), g(f(\bar{x})))$ (decode-vs-decode, relative to the model's own reconstruction of the mix) and SI-SDR$^{\mathrm{gt}}_{\mathrm{lin}}$ = $\mathrm{SISDR}(g(\bar{z}), \bar{x})$ (vs. ground truth, comparable across models). We also report MixRate = $\mathcal{L}_{\mathrm{mix}}(g(\bar{z}), \bar{x}) / \mathcal{L}_{\mathrm{mix}}(g(f(\bar{x})), \bar{x})$ with $\mathcal{L}_{\mathrm{mix}}$ the three-term loss of Eq. 2 in both terms, and the latent error $r = \bar{z} - f(\bar{x})$. The ℓ_{lat} column of Table I is the customary ratio $\|r\|^2 / \|f(\bar{x})\|^2$, which Section IV-B shows to be *translation-sensitive*; every latent quantity is a per-sample mean over the evaluated units. SI-SDR fits a gain before scoring, so we report that gain separately, on 3,000 samples. FAD uses music-finetuned LAION-CLAP embeddings of non-overlapping 10 s windows at 48 kHz, not VGGish, so values are not comparable with VGGish FAD.

SI-SDR$_{\mathrm{lin}}$ shares the decode-vs-decode structure of prior additivity metrics [5] and is additionally sensitive to decoder sampling. A consistency-model decoder [12] draws fresh noise per call: decoding the *same* latent twice is deterministic under shared noise but scores -1.8 dB under independent noise, with no latent arithmetic performed. On identical pairs the same checkpoints score -9.1 and -7.3 dB when the two decodes draw independent noise and $+3.6$ and $+5.9$ dB when they share it, a 13 dB swing driven by the protocol rather than the model; which protocol prior work used is not stated in enough detail for us to determine. We treat SI-SDR$^{\mathrm{gt}}_{\mathrm{lin}}$ and MixRate as the metrics of record and share decode noise everywhere.

D. Experimental setup

The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [8], combined multi-scale-STFT and multi-period discriminator) at 44.1 kHz on 1-second mono chunks, trained on a $\sim$950-hour union of FMA-large, MAESTRO and MUSDB18-HQ [13].[1] Two compression regimes are used: 7.66$\times$ (d=128) and 15.3$\times$ (d=64). Mixing losses are added either as a 25k-step fine-tune from a shared converged baseline or from scratch over 250k steps, each against a matched control that differs only in the mixing terms. **Checkpoint selection.** The released weights are the checkpoint with the lowest validation loss, and the validation loader read the test split; for every fine-tune this coincides with the final step, while both from-scratch runs were selected at step 240k of 250k. This is why Section IV-D repeats the key measurement on a corpus that took no part in selection.

IV. Results

A. Mixing supervision improves ground-truth mixing

Table I compares same-start fine-tunes at 7.66$\times$. On the ground-truth metric, $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ improves SI-SDR$^{\mathrm{gt}}_{\mathrm{lin}}$ from 8.0 to 10.2 dB and is the only configuration with MixRate < 1 (0.97). Reconstruction is similar to the matched control at this rate, SDR$_{\mathrm{rec}}$ spanning 11.2–11.4 dB and FAD 0.044–0.046; these are single runs, so an absence this small is not established beyond seed variance.

The improvement is confirmed on audio that took no part in training or in checkpoint selection. On 240 held-out MoisesDB recordings [14], $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ gains $+1.73$ dB $[+1.69, +1.78]$ at 7.66$\times$ and the combined recipe $+1.33$ $[+1.27, +1.39]$ at 15.3$\times$, for reconstruction changes of $+0.04$ and -0.07 dB. Intervals resample recordings; mixing pairs join two chunks,

[1]Code, training configurations, pretrained weights, every metric reported here, and audio examples: https://github.com/nurdauletakhanov/musicgen

Loss	$\mathrm{SDR}_{\mathrm{rec}}$	$\mathrm{SDR}_{\mathrm{lin}}$	$\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$	ℓ_{lat}	MixRate	FAD
baseline (no mix)	+11.3	+12.0	+8.0	0.068	1.203	0.045
$\mathcal{L}_{\mathrm{dec}}$	+11.4	+12.1	+10.2	0.052	0.972	0.044
$\mathcal{L}_{\mathrm{dec}}$ + disc-on-mix	+11.4	+16.0	+9.9	0.050	1.052	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=5)	+11.3	+12.8	+8.4	0.044	1.175	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=10)	+11.3	+13.1	+8.5	0.039	1.165	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=20)	+11.3	+13.5	+8.7	0.034	1.153	0.045
$\mathcal{L}_{\mathrm{dec}}$ frozen-enc	+11.2	+11.7	+9.5	0.072	1.056	0.046

TABLE I

SAME-START FINE-TUNING COMPARISON (7.66×, FULL TEST SET, α=0.5). $\mathrm{SDR}_{\mathrm{lin}}$ IS DECODE-VS-DECODE; $\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ IS VS. GROUND TRUTH. SINGLE RUN EACH.

about 80% of them from the same recording. A pair crossing two recordings enters a resample $m_i m_j$ times and a pair inside one recording m_i times, where m is that recording's multiplicity; weighting both as $m_i m_j$ would overweight the within-recording majority. The gain also holds across α: swept under one protocol against its matched control, the $\text{SI-SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ swing is 1.9 dB with $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ against 3.2 dB without, and $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ leads at all five coefficients.

B. What the training comparisons establish

The encoder-only variant $\mathcal{L}^{\mathrm{enc}}_{\mathrm{mix}}$ leaves MixRate high (1.15–1.18): it constrains the encoder without making the decoder equivariant. Training only the decoder under $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$, with the encoder frozen, retains most of the gain (9.5 dB against 10.2 joint and 8.0 baseline). Joint training is best among the tested configurations. Decoder adaptation is therefore largely sufficient here, which does not show it is necessary or that the decoder is the unique locus; a decoder-frozen counterpart was not trained. **Encoder geometry has no normalization-independent answer.** The customary ratio $\|r\|^2/\|f(\bar{x})\|^2$ is translation-sensitive. The coefficients of $r = \bar{z} - f(\bar{x})$ sum to zero, so under the relabelling of Section III-B the numerator is unchanged while a large enough c inflates the denominator and drives the ratio toward zero. Table II re-measures the three contrasts under four normalizations on the held-out corpus. The unnormalized residual and the ratio to the endpoint distance, both translation-invariant, favour the mixing models throughout; the ratio to each encoder's distance from its own $f(0)$, also translation-invariant, reverses at 7.66× and is indistinguishable from zero at 15.3×. The controls carry two to three times the offset norm ($\|f(0)\|$ 18.1 and 14.8 against 5.4–6.7), which inflates that denominator for them. Only the endpoint ratio is additionally invariant to rescaling $f \to sf$ with the decoder adjusted; the unnormalized residual is not. We therefore make no claim about improved encoder geometry.

C. Decoder consistency can misrepresent mixing quality

Adding an adversarial term to the mixed path does *not* improve ground-truth equivariance (10.2 → 9.9 dB, MixRate 0.97 → 1.05), yet it raises the decode-vs-decode $\text{SI-SDR}_{\mathrm{lin}}$ from 12.1 to 16.0 dB. The held-out corpus confirms the direction and sharpens it: the adversarial term *costs* 0.49 dB

normalization of $r = \bar{z} - f(\bar{x})$	$\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$	+disc	15.3×
$\|r\|^2/\|f(\bar{x})\|^2$ *(transl.-sens.)*	-29.9±1.7	-36.0±1.9	-45.1±2.8
$\|r\|^2/D$	-2.3±0.1	-2.6±0.2	-3.0±0.4
$\|r\|^2/\|f(x_1)-f(x_2)\|^2$	-16.7±0.8	-20.3±0.9	-30.1±1.4
$\|r\|^2/\|f(\bar{x})-f(0)\|^2$	+13.4±1.0	+8.3±1.2	-0.2±2.1†

TABLE II

LATENT ERROR UNDER FOUR NORMALIZATIONS, HELD-OUT CORPUS: PAIRED DIFFERENCE (TREATMENT − CONTROL, $\times 10^{-3}$, NEGATIVE IS BETTER) ± HALF THE 95% INTERVAL FROM A DYADIC BOOTSTRAP OVER 240 RECORDINGS. † MARKS THE ONE INTERVAL CONTAINING ZERO.

Model	drums	bass	vocals	other	all	gap↓
7.66× no mix	+4.2	+0.4	+5.2	+4.0	+3.4	5.2
$+f(0)$	+8.8	+6.8	+9.6	+8.0	+8.3	0.4
7.66× $+\mathcal{L}_{\mathrm{dec}}$	+5.7	+4.9	+7.3	+5.2	+5.8	2.9
$+f(0)$	+8.5	+6.9	+9.4	+8.3	+8.3	0.5
7.66× $+\mathcal{L}_{\mathrm{dec}}$+disc	+6.2	+5.4	+7.5	+5.2	+6.1	2.6
$+f(0)$	+8.5	+7.0	+9.3	+8.1	+8.2	0.4
15.3× no mix	+1.2	-1.6	+3.0	+2.2	+1.2	6.6
$+f(0)$	+7.3	+5.8	+8.8	+7.2	+7.3	0.5
15.3× $+\mathcal{L}_{\mathrm{dec}}$+disc	+5.4	+4.7	+6.9	+4.0	+5.2	2.4
$+f(0)$	+7.3	+6.1	+8.5	+7.2	+7.3	0.4
M2L +mix	-9.9	-12.3	-9.9	-12.7	-11.2	8.4

TABLE III

STEM REMOVAL VIA LATENT SUBTRACTION (SI-SDR, DB; MUSDB18 TEST). "GAP" IS THE LINEARITY TAX AGAINST EACH MODEL'S OWN ENCODE–DECODE REFERENCE (LOWER IS BETTER). "$+f(0)$" ROWS READ THE SAME CHECKPOINT IN RECENTERED COORDINATES. AGGREGATES OVER 5,880 UNITS FROM 49 TRACKS.

$[-0.51, -0.48]$ of ground-truth equivariance while adding +4.06 dB of decode-consistency. The two measurements therefore rank the same checkpoints differently and cannot serve as interchangeable evidence of mixing capability. We do not identify the mechanism; one candidate is that both decodes are driven onto a common realistic manifold. We do not separately report ground-truth SI-SDR for the mixed-input reconstruction route $g(f(\bar{x}))$, though MixRate scores it against $\bar{x}$ under the mixed-path loss; and whether the discriminator acts alone is untested.

D. Subtraction depends strongly on the latent origin

We remove a stem by latent subtraction and measure SI-SDR against the true residual on 49 of the 50 MUSDB18 test tracks (one is excluded because its stems sum to the mixture at only 18.5 dB). This is an *oracle* setting: the removed stem is encoded, so it tests arithmetic, not blind separation. By Section III-B we also report the recentered form $g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}) + f(0))$.

Raw subtraction favours the mixing loss; recentering substantially reduces the difference. Raw, $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ raises subtraction quality at 7.66× from +3.4 to +5.8 dB, and the matched 15.3× pair from +1.2 to +5.2. Adding $f(0)$ gains +2.0 to +6.0 dB for every evaluated configuration, leaving each within about $0.5\,\mathrm{dB}$ of its own reconstruction reference in aggregate; per track the tail is wider. The between-model difference then falls to $-0.03\,\mathrm{dB}$ (95% track-cluster CI $[-0.15, +0.10]$) and $+0.01$ ($[-0.19, +0.21]$).

Because these checkpoints were selected on this same split (Section III-D), we repeat the intervention on MoisesDB, 240 recordings absent from training and from selection, under a protocol registered before the audio was downloaded. Raw subtraction again favours the mixing models by $+2.3$ to $+4.5\,\mathrm{dB}$ and recentering removes 87–95% of that, shrinking the gap on 229–236 of the 240 recordings; on the rest it overshoots and reverses the sign. The residue is resolvable there ($+0.29$, $+0.12$ and $+0.42\,\mathrm{dB}$, intervals excluding zero), so a small training-associated advantage survives correction. Units are scored where the removed stem is active, at least $-50\,\mathrm{dBFS}$ and within $30\,\mathrm{dB}$ of the mixture energy (24,648 of 28,800 kept; vocals lose the most, 70.4%). The rule excludes silent *targets* and does not make the SI-SDR reference nonsilent, that reference being the remaining mixture. The protocol was committed before the audio was downloaded and is released as `HOLDOUT_PROTOCOL.md` with the checkpoint hashes the evaluation verifies at run time.

Baselines without latent arithmetic. The unchanged mixture scores $+17.1\,\mathrm{dB}$ against the residual, above every model's reconstruction reference, since the removed stem is a small share of the energy: the metric certifies arithmetic relative to that reference, not removal. Recentered latent subtraction beats decode-then-subtract by $+0.7$ to $+0.9\,\mathrm{dB}$; raw subtraction loses to it everywhere.

E. Scope

Table IV extends the comparison. **Compression:** trained from scratch at $15.3\times$ against a matched control, the combined $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$+disc recipe improves SI-SDR$_{\mathrm{lin}}^{\mathrm{gt}}$ from 6.8 to 8.6 dB and MixRate from 1.19 to 1.05 at equal reconstruction, while FAD rises from 0.052 to 0.061; whether that is perceptible is untested. That pair differs by the whole recipe and, unlike the fine-tunes, does not share initial weights, so it is an observation about the combined recipe in those two runs rather than an isolated effect of $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$. The gain also appears on all three test domains at both rates ($+1.2$ to $+3.2\,\mathrm{dB}$). **Level:** the decoded mix is $1.6\,\mathrm{dB}$ too quiet without the loss and $0.9\,\mathrm{dB}$ with it, against a reconstruction level error of $0.8\,\mathrm{dB}$, so the loss reduces level error as well as spectral error. **Architecture:** we examine whether the benefit extends to a different autoencoder family under the tested fine-tuning protocol, applying the same mixed-latent supervision to Music2Latent [12] under its native consistency loss. Mixing SI-SDR improves against a continued-training control ($-9.5 \rightarrow -7.8\,\mathrm{dB}$), which itself moves equivariance by $< 0.2\,\mathrm{dB}$, while absolute waveform performance remains poor. The systems differ in too many respects for this to isolate a cause, and we claim nothing about consistency autoencoders in general.

Model	SDR$_{\mathrm{rec}}$	SDR$_{\mathrm{lin}}$	SDR$_{\mathrm{lin}}^{\mathrm{gt}}$	MixRate	FAD
GAN, 7.66×, no mix	+11.3	+12.0	+8.0	1.203	0.045
GAN, 7.66×, +$\mathcal{L}_{\mathrm{dec}}$	+11.4	+12.1	+10.2	0.972	0.044
GAN, 15.3×, no mix	+10.0	+10.6	+6.8	1.187	0.052
GAN, 15.3×, +$\mathcal{L}_{\mathrm{dec}}$+disc	+9.9	+14.6	+8.6	1.046	0.061
M2L, no mix (baseline)	-3.3	+5.0	-9.6	1.101	0.127
M2L, fine-tune control	-3.1	+5.1	-9.5	1.125	0.134
M2L, +mix	-2.6	+6.9	-7.8	1.086	0.119

TABLE IV

COMPARISON ACROSS COMPRESSION AND ARCHITECTURE (FULL TEST SET, α=0.5, EMA WEIGHTS FOR MUSIC2LATENT). THE M2L "FINE-TUNE CONTROL" IS THE CONTINUED-TRAINING BASELINE WITH MIXING DISABLED.

V. CONCLUSION

Explicit mixing supervision improves ground-truth waveform mixing in the evaluated GAN autoencoders, with little change in reconstruction SI-SDR, and the improvement holds on a held-out corpus that took no part in training or in checkpoint selection. Decoder consistency and stem subtraction, however, report different properties of the same checkpoints: an adversarial term raises decode-consistency while lowering ground-truth mixing, and most of the raw subtraction advantage disappears when each checkpoint is recentered at its silence latent. The three measurements are therefore not interchangeable evidence of mixing capability. **Limitations.** Comparisons use single training runs; fine-tunes share a starting checkpoint, while the from-scratch pair shares a configuration but not initial weights, as training sets no explicit seed. Intervals quantify recording-level evaluation variation conditional on the released checkpoints, not variation across training runs. Checkpoint selection used the test split; audio is mono and untested by listeners.

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